

☐ د ٤

☐ هـ ٤

6

* رقم الكمبيوتر

2

7

* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

DJQX

 Will be reviewed

8

How many triangles are drawn if we use the drawing mode GL_TRIANGLE_STRIP with six vertices between glBegin(GL_TRIANGLE_STRIP) and glEnd()?

☐ 2

☐ 3

☒ 4

☐ 5

 Will be reviewed

Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(90.0, 0, 0, 1.0);
glPushMatrix();
glTranslatef(-1.0f, -2.0f, -3.0f);
glRotatef(-90.0, 0, 0, 1.0);
glScalef (-1.0,0.5, -1.0);
glPopMatrix();
glRotatef(90.0, 0, 0, 1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☐ A

☐ B

☒ C

☐ D

 Will be reviewed

10

Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(90.0, 0, 0, 1.0);
glScalef(-1.0, 2.0, -1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☐ A

☒ B

☐ C

☐ D

 **Will be reviewed**

11

This 3D homogeneous point: (4,3,2,1) corresponds to what Cartesian point:

☐ (4,3,2,1)

☐ (3,2,1)

☐ (3/4, 1/2, 1/4)

☒ (4,3,2)

 **Will be reviewed**

12

The Modelview matrix after executing the following OpenGL code segment is:
`glMatrixMode(GL_MODELVIEW); glLoadIdentity(); glRotatef(45.0, 0, 0, 1.0);
glRotatef(45.0, 0, 0, 1.0);`

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☒ A

☐ B

☐ C

☐ D

 **Will be reviewed**

13

The following two code segments are equivalent:

Code#1

```
glTranslatef(-1.0f, -2.0f, -3.0f);  
glRotatef(-90.0, 0, 0, 1.0);  
dawShape_3();
```

Code#2

```
glRotatef(-90.0, 0, 0, 1.0);  
glTranslatef(-1.0f, -2.0f, -3.0f);  
dawShape_3();
```

☐ True

☒ False

 **Will be reviewed**

14

Generally, what primitive polygon is used for creating a mesh to represent a complex object?

☒ Triangle

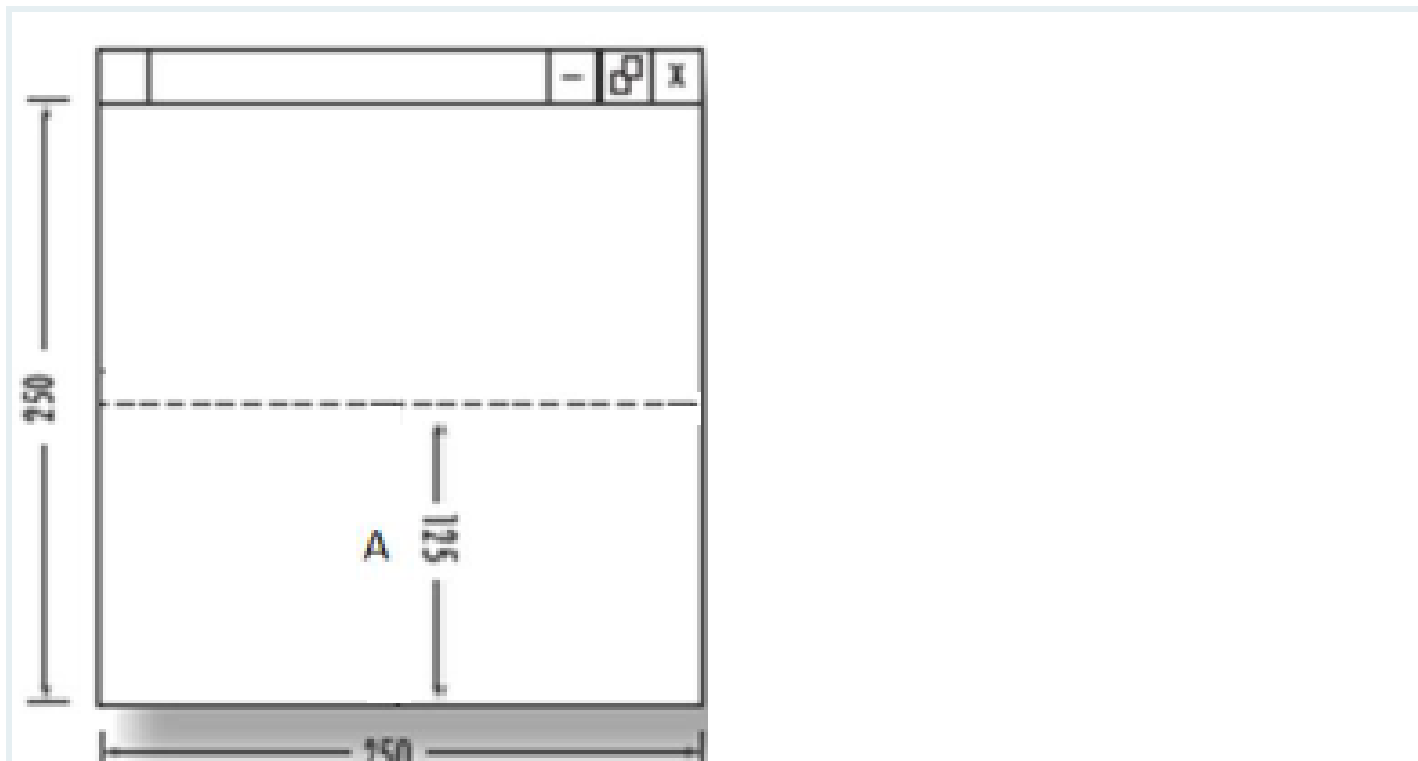
☐ Cube

☐ Sphere

☐ Circle

 **Will be reviewed**

To map the output of the OpenGL code to the area A (as shown in the figure), we use the.



- ☒ `glViewport(0, 0, 250, 125);`
- ☐ `glViewport(0, 0, 125, 250);`
- ☐ `glViewport(125, 125, 125, 125);`
- ☐ `glViewport(250, 125, 0, 0);`
- ☐ `glViewport(0, 250, 0, 125);`

 **Will be reviewed**

OpenGL can draw convex polygons.

- ☒ True

☐ False

 Will be reviewed

17

Rotate (z, 90) * Rotate (z, 45) = Rotate (z, 45) * Rotate (z, 90)

☒ True

☐ False

 Will be reviewed

18

The OpenGL represents the following matrix as

$$m = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

☐ GLfloat m[] = { 1.0f, 5.0f, 1.0f, 2.0f, 6.0f, 2.0f, 3.0f, 7.0f, 3.0f, 4.0f, 8.0f, 4.0f};

☐ GLfloat m[] = { 1.0f, 2.0f, 3.0f, 4.0f, 5.0f, 6.0f, 7.0f, 8.0f, 1.0f, 2.0f, 3.0f, 4.0f};

☐ GLfloat m[] = { 1.0f, 2.0f, 3.0f, 4.0f, 5.0f, 6.0f, 7.0f, 8.0f, 1.0f, 2.0f, 3.0f, 4.0f, 0.0f, 0.0f, 0.0f, 1.0f};

☒ GLfloat m[] = { 1.0f, 5.0f, 1.0f, 0.0f, 2.0f, 6.0f, 2.0f, 0.0f, 3.0f, 7.0f, 3.0f, 0.0f, 4.0f, 8.0f, 4.0f, 1.0f};

 Will be reviewed

19

What does glClear do?

- ☒ It clears all OpenGL buffers
- ☐ It clears the OpenGL buffers specified in an argument
- ☐ It sets the window color to black
- ☐ None of the above

 **Will be reviewed**

20

Which of the following is not a valid argument to glBegin ()?

- ☐ GL_POINTS
- ☐ GL_LINES
- ☐ GL_LINE_STRIP
- ☐ GL_TRIANGLES
- ☒ GL_QUADS_FAN

 **Will be reviewed**

21

glOrtho is used to define a _____ matrix.

- ☐ modelview
- ☒ projection

- ☐ view
- ☐ model

 **Will be reviewed**

22

What is the significance of i,f,v,d... in function naming conventions found in OpenGL?

- ☐ Specify the target type for variables written by OpenGL on the CPU-side only
- ☐ OpenGL's naming conventions vary and must be examined in the specification for the version you are using
- ☒ Specify the expected parameter types (int, float, pointer, double, etc.)
- ☐ Specify the output target (immediate mode, frame buffer, virtual mode, double-buffer, etc.)

 **Will be reviewed**

23

Rotating the 3D point (0,-1,0) by -90 around z-axis gives what:

- ☐ (1,0,0)
- ☒ (-1,0,0)
- ☐ (0,1,0)
- ☐ (0,-1,0)
- ☐ (-1,-1,0)

 Will be reviewed

24

Consider the following matrix.

This matrix achieves the following in the global coordinate system: (Note: R_z means rotation around the z axis and S_z means scale in the z direction)

0.707	0.707	0	0
-0.707	0.707	0	0
0	0	2	0
0	0	0	1

- ☐ R_z of 45° followed by S_z of 2
- ☐ R_y of 45° followed by S_z of 2
- ☒ S_z of 2 followed by R_z of -45°
- ☐ S_z of 2 followed by R_y of 45°

 Will be reviewed

25

The plane N is given by $2x+2y+2z = 6$ What is the normal to the plane N ?

- ☐ (1, 1, 0)
- ☒ (1, 1, 1)
- ☐ (0, 1, 0)
- ☐ (1, 0, 1)

 Will be reviewed

Is the following code correct?

```
glBegin (GL_QUADS);  
glVertex2i (1, 0);  
glVertex2i (1, 1);  
glVertex2i (-1, 1);  
glVertex2i (-1, 0);  
glEnd (GL_QUADS);
```

☐ Yes

☒ No

 **Will be reviewed**

Which of the following is not a valid argument to glMatrixMode?

☐ GL_PROJECTION

☒ GL_DEPTH

☐ GL_MODELVIEW



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